ALLEN TU

Ph.D. Student, Computer Science — University of Maryland

Research Areas: Reliable and Scalable Vision Systems, 3D/4D Reconstruction, Biometrics, Generative Models

Email: atu1@umd.edu | **Website**: tuallen.github.io

LinkedIn: linkedin.com/in/allentu | **Google Scholar**: Allen Tu

SELECTED PUBLICATIONS

1. **Allen Tu***, H. Ying*, A. Hanson, Y. Lee, T. Goldstein, and M. Zwicker, 'SpeeDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping'. *CVPR*, 2026.
2. P. Asthana, A. Hanson, **Allen Tu**, T. Goldstein, M. Zwicker, and A. Varshney, 'SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting'. *CVPR*, 2026.
3. **Allen Tu**, K. Narayan, J. Gleason, J. Xu, M. Meyn, and V. Patel, 'TransFIRA: Transfer Learning for Face Image Recognizability Assessment'. *FG*, 2026.
4. A. Hanson*, **Allen Tu***, V. Singla, M. Jayawardhana, M. Zwicker, and T. Goldstein, 'PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting'. *CVPR*, 2025.
5. A. Hanson, **Allen Tu**, G. Lin, V. Singla, M. Zwicker, and T. Goldstein, 'Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives'. *CVPR*, 2025.

* denotes equal contribution.

EDUCATION

University of Maryland, College Park — College Park, MD
Ph.D. Student, Computer Science — *January 2025 – May 2028*

- Advised by Professor Tom Goldstein; collaborating with Professor Matthias Zwicker

B.S./M.S. in Computer Science, Minor in Statistics — *August 2019 – December 2024*

RESEARCH EXPERIENCE

Amazon — Bellevue, WA
Incoming Applied Science Intern — *June 2026 – October 2026*

- Joining the Last Hundred Yards team to research 3D reconstruction for robotics

University of Maryland Institute of Advanced Computer Studies — College Park, MD
Graduate Research Assistant — *August 2023 – Present*

- Developing scalable 3D reconstruction pipelines deployed for unconstrained novel-view synthesis in real-world environments under the IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA) program
- Created pruning, rasterization, and motion distillation methods that accelerate static and dynamic 3D Gaussian Splatting, producing representations with over $10\times$ fewer primitives while preserving visual fidelity [1, 4, 5]
- Incorporated uncertainty-aware diffusion and super-resolution priors into 3D reconstruction pipelines to improve fidelity under sparse and low-resolution supervision while mitigating artifacts and cross-view inconsistencies [2]

Systems & Technology Research — Arlington, VA
Computer Vision Research Intern — *June 2022 – January 2026*

- Engineered multimodal biometric systems fusing face, body, and gait recognition for robust identification under severe conditions in the IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR) program
- Introduced an explainable transfer-learning method for face image quality assessment (FIQA), enabling recognizability-aware probe filtering and weighted template aggregation for face and body recognition [3]
- Demonstrated state-of-the-art template-based recognition on the proprietary BRIAR surveillance benchmark, improving TAR from 0.43 to 0.87 at 1e-3 FMR using an encoder trained solely on public data [3]

SELECTED AWARDS AND SERVICE

FG 2026 Doctoral Consortium

Workshop Organizer: SPAR-3D: Security, Privacy, & Adversarial Robustness in 3D Generative Vision Models, CVPR 2026